Name: Sheik Ismail Ajmal
Course: MCA Cybersecurity

Password Strength Analyzer and Breach Checker

Java Project Report

This report presents the development and implementation of a Password Strength Analyzer and Breach Checker application developed using Java Swing and SQLite database. The project focuses on password security analysis, breach detection, and cybersecurity awareness.

Abstract

Passwords are one of the most commonly used authentication mechanisms in modern digital systems. Weak passwords can easily be guessed, cracked, or exploited by attackers using techniques such as brute force attacks, dictionary attacks, and credential stuffing. Many users continue using passwords that have already been exposed in previous data breaches. The Password Strength Analyzer and Breach Checker project is developed to help users understand password security and improve their password practices. The application evaluates password strength using factors such as password length, uppercase characters, lowercase characters, numbers, and special symbols. The system also checks whether the entered password exists in a breached password database. The project is implemented using Java Swing for the user interface and SQLite for database management. The system stores password analysis logs for monitoring and educational purposes.

Introduction

In today's digital world, passwords are essential for securing online accounts, applications, databases, and systems. However, many users create weak passwords that are easy to guess or reuse passwords across multiple platforms. This creates serious cybersecurity risks and increases the possibility of unauthorized access. Cybercriminals often use leaked password datasets to perform attacks such as credential stuffing and account takeover. Passwords like '123456', 'password', and 'qwerty' are among the most commonly breached passwords globally. This project helps users identify weak and breached passwords by analyzing password complexity and checking against a local breached password dataset. The application also provides suggestions to improve password quality.

Objectives

• Analyze password strength using security rules.
• Detect passwords available in breach datasets.
• Provide recommendations for stronger passwords.
• Store password analysis results into database logs.
• Create a user-friendly Java application.
• Demonstrate practical cybersecurity concepts.

System Modules

User Interface Module

The User Interface module is developed using Java Swing. It allows users to enter passwords, view password strength, check breach status, and receive security suggestions.

Password Strength Checker

This module analyzes password complexity based on length, uppercase letters, lowercase letters, numbers, and special characters. A score is generated to classify the password as Weak, Medium, or Strong.

Breach Checker Module

The breach checker searches the SQLite database for passwords that exist in the breached password dataset. If the password is found, the system displays a warning message.

Logging Module

The logging module stores password analysis results including password, strength, and breach status into the database for monitoring purposes.

Database Connection Module

This module establishes and manages SQLite database connectivity using JDBC driver.

Technologies and Database Design

The project is developed using Java programming language. Java Swing is used for building the graphical user interface while SQLite is used as the backend database. JDBC driver is used for connecting Java with SQLite database. The development environment used for this project is IntelliJ IDEA.

Table: breached_passwords	
Field Name	Data Type
id	INTEGER
password	TEXT

Table: logs	
Field Name	Data Type
id	INTEGER
password	TEXT
strength	TEXT
breached	INTEGER

Implementation

The project implementation started with setting up SQLite database connectivity using JDBC. Two database tables were created for storing breached passwords and analysis logs. The Password Strength Checker module was implemented using Java conditions and regular expressions. The application checks password properties such as length, uppercase letters, lowercase letters, numbers, and special symbols. The Breach Checker module compares user-entered passwords with breached password records stored in the database. PreparedStatement was used to prevent SQL injection vulnerabilities. The User Interface was designed using Java Swing components such as JFrame, JButton, JLabel, JCheckBox, and JPasswordField. Additional features such as password suggestions, show/hide password option, and color indicators were also implemented.

Testing and Results

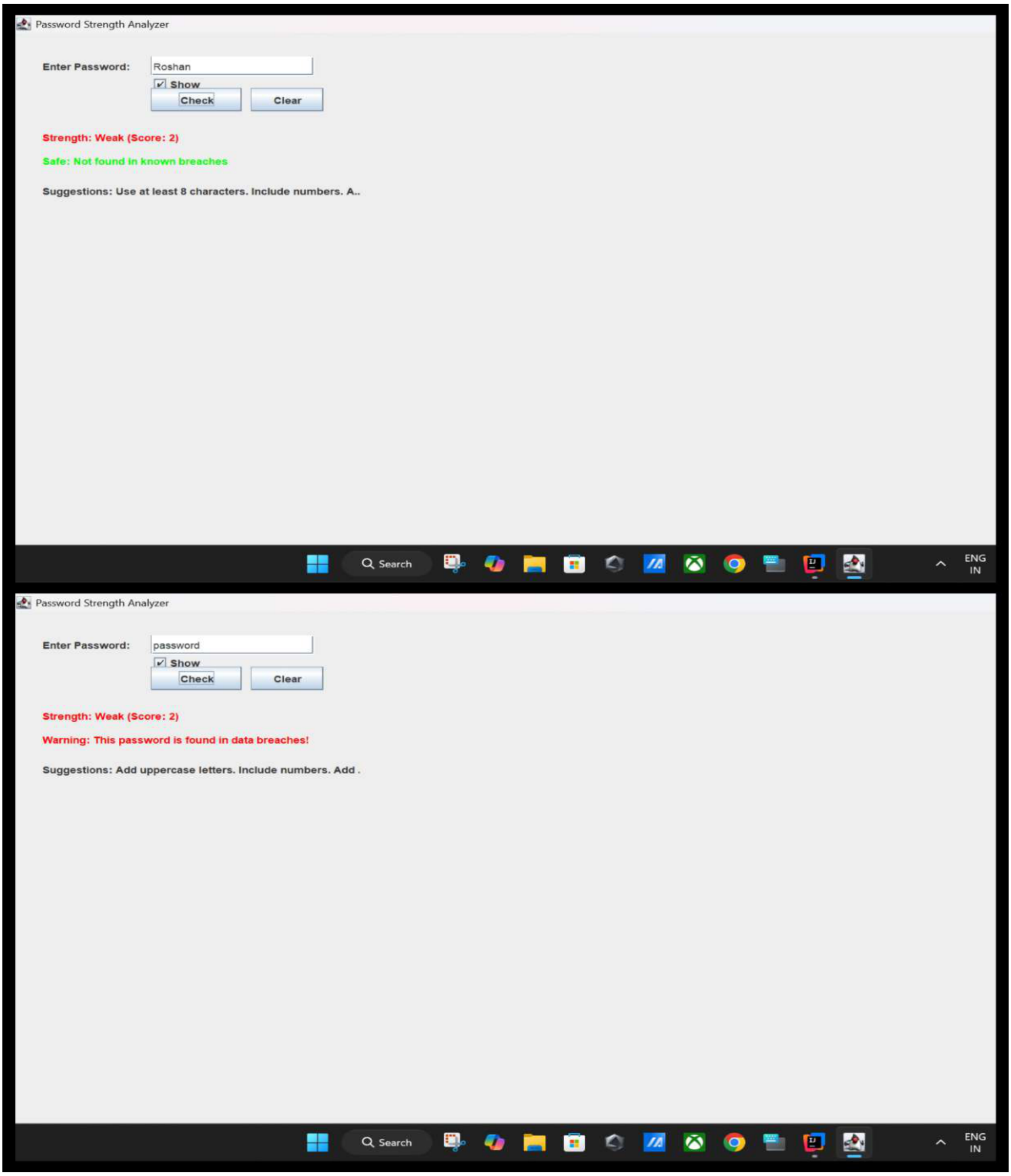
The project was tested using multiple password combinations to verify strength analysis and breach detection accuracy. Functional testing was performed to ensure proper operation of the user interface and database logging.

Password	Strength	Breached
123	Weak	No
password	Weak	Yes
Admin123	Medium	No
Admin@123	Strong	No
123456	Weak	Yes

The application successfully identified weak passwords and detected passwords available in the breached password dataset. The logging module also stored password analysis records correctly in the SQLite database.

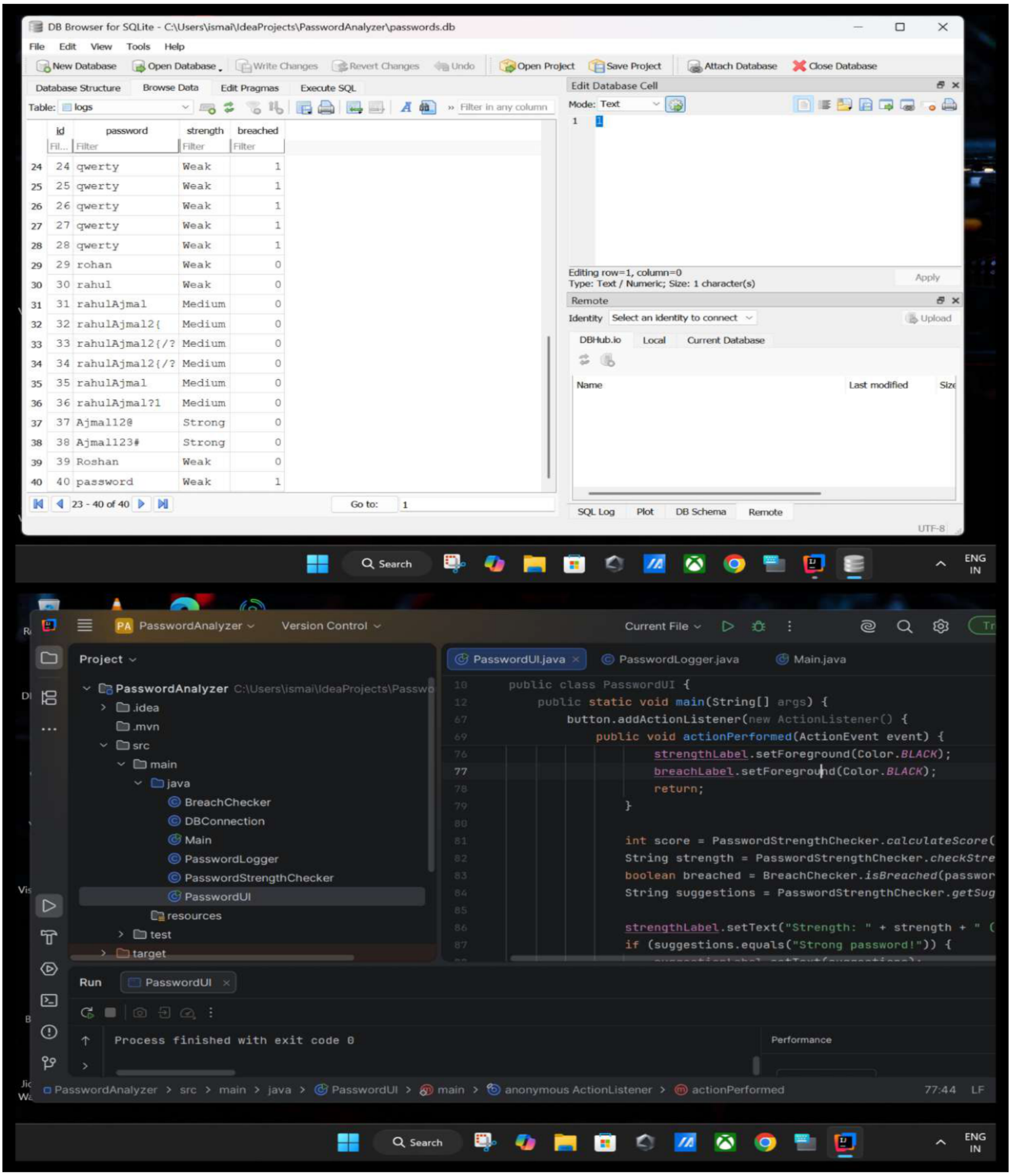
Project Screenshot - User Interface

The following screenshots show the Password Strength Analyzer application interface. The application displays password strength, breach status, and suggestions for improving password quality.



Project Screenshot - Database and Source Code

The screenshots below show the SQLite database logs and the Java source code structure implemented in IntelliJ IDEA. The logs table stores password analysis records while the Java project contains modules for UI, logging, breach checking, and password analysis.



Advantages, Limitations and Conclusion

Advantages:

• User-friendly interface

• Lightweight application

• Offline functionality

• Detects breached passwords

• Improves password awareness

Limitations:

• Uses local breach dataset only

• No online API integration

• Basic rule-based password analysis

Conclusion:

The Password Strength Analyzer and Breach Checker project was successfully developed using Java Swing and SQLite. The application demonstrates practical cybersecurity concepts such as password analysis, breach detection, database integration, and secure query handling. The project helps users understand the importance of strong passwords and encourages safer password practices. The application can be further enhanced with advanced security features and cloud-based integrations in the future.

References:

1. Oracle Java Documentation

2. SQLite Documentation

3. JDBC Documentation

4. OWASP Password Security Guidelines